Chapter 1

Multiplex Systemics for Wholeness-Making: Toward a More Holistic Philosophy of Educational Technology

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Abstract

As artificial intelligence reshapes education, ethical discourse has focused primarily on addressing bias, privacy, and transparency. These are essential concerns, but they operate within narrow technocratic frameworks. This chapter introduces *multiplex systemics*, a comprehensive philosophical approach that broadens the ethical horizon by recognizing reality's layered nature, the web of dialectical tensions shaping educational systems, and the imperative of *wholeness-making*: the marshaling of diverse potentials toward normative ends. Drawing on complexity science, Islamic philosophical concepts of *kamāl* (fullness of being), and both-and thinking, we propose a framework that evaluates educational technology not merely by instrumental effectiveness but by its contribution to human flourishing across physical, psychological, social, and spiritual dimensions. Through detailed analysis of AI writing assistants in higher education, we demonstrate how multiplex systemics reveals hidden costs and generative possibilities obscured by reductive either-or thinking. This work continues and extends our ongoing scholarship on applied multiplexity, offering educators, technologists, and policymakers a coherent vision for designing AI-enhanced education that serves genuine learning, human dignity, and civilizational flourishing rather than narrow optimization.

1.1 Introduction: Beyond Technocratic AI Ethics

1.1.1 The Current State of AI Ethics in Education

The integration of artificial intelligence into educational systems has prompted necessary ethical scrutiny (Holmes et al., 2022). Current discussions predominantly address what we might characterize as *first-order ethical concerns*: algorithmic bias in adaptive learning platforms, student data privacy, transparency in automated grading, and accountability when AI systems make consequential decisions about learners. These concerns demand sustained attention from educators, technologists, and policymakers, and have generated substantial scholarship on fairness, accountability, and transparency in educational AI.

However, this prevailing discourse operates within fundamentally limited parameters. As Holmes et al. (2025) argue in their recent manifesto on critical studies of AI in education, the technocratic framing treats ethical challenges as *problems to be solved* through better algorithms, stronger regulations, or more transparent systems. This instrumental rationality, what Postman (1993) called "technopoly," assumes technology is essentially neutral, with ethical issues arising only from poor implementation or malicious use.

Such narrow framing obscures deeper questions that must be asked: What conception of the human person underlies our educational AI systems? What vision of society and human flourishing guides our technological choices? What are we optimizing for, and who decides? As Birhane (2021) observes, algorithmic systems embody and operationalize particular ontological assumptions about human nature, often reducing persons to data points and predictive patterns. The resulting "flattening" of human complexity into computationally tractable variables represents not merely a technical limitation but a philosophical impoverishment with profound ethical implications.

We can identify three overlapping approaches in AI ethics scholarship: *computational approaches* focus on technical solutions like fairness-aware algorithms and privacy-preserving methods; *philosophical approaches* examine foundational questions about autonomy, dignity, and justice; and *critical approaches* interrogate power relations, structural inequities, and the political economy of AI systems (Qadir and Senturk, 2026a). While each contributes valuable insights, critical approaches hold the most promise for genuinely transformative change, particularly for marginalized communities disproportionately impacted by algorithmic systems. Yet even critical AI ethics often lacks a comprehensive philosophical anthropology, a coherent account of human nature, development, and flourishing that could ground normative judgments about technology's proper role in education.

Song (2024) argues compellingly that education must move beyond narrow "intelligence" paradigms that dominate both AI development and educational technology discourse. The fixation on measurable cognitive performance, while capturing something real, systematically excludes dimensions essential to human flourishing—wisdom, character, relational capacity, spiritual depth. This critique aligns with our multiplex systemics framework, which insists that educational technology must be evaluated across multiple irreducible levels rather than reduced to optimization of intelligence metrics.

1.1.2 The Need for Multiplex Thinking

The limitations of current AI ethics discourse reflect a broader intellectual tendency we characterize as *uniplex thinking*—the reduction of complex realities to single dimensions, the flattening of layered truths into binary choices, and the privileging of quantifiable metrics over qualitative depth. This reductionist paradigm, rooted in classical Western logic's law of excluded middle, has yielded tremendous analytical power. Yet it fundamentally misrepresents educational phenomena, which are inherently *multi-layered* (involving physical, psychological, social, cultural, and spiritual dimensions), *relational* (emerging from webs of interaction rather than isolated individuals), *contextual* (shaped by historical and situational factors), *teleological* (oriented toward purposes transcending mere optimization), and *dialectical* (characterized by creative tensions between complementary opposites).

Consider a seemingly straightforward question: Should we use AI tutoring systems in mathematics education? A uniplex approach frames this as an either-or choice, weighing efficiency gains against potential harms. But the reality is far more nuanced. AI tutoring is simultaneously a technical tool that can personalize instruction, a pedagogical relationship mediating human connection, an ideological statement reflecting particular values about learning, a practice shaping how knowledge is acquired, and a phenomenological experience affecting learner identity. Understanding and ethically navigating AI in education requires what we call *multiplex systemics*—a framework recognizing multiple layers of reality, attending to systemic interdependencies, and aiming toward wholeness-making rather than mere problem-solving.

This work builds directly on our ongoing scholarship developing and applying multiplexity across domains. In previous work, we have articulated multiplexity as a multi-layered understanding of reality, knowledge, and truth that supports both-and thinking instead of either-or logic (Şentürk, 2012, 2025). We have demonstrated its value for engineering education (Qadir, 2022), AI literacy (Qadir and Senturk, 2026a), and healthcare (Qadir et al., 2025). This chapter extends our applied multiplexity program by developing *multiplex systemics* specifically for educational technology ethics. Where our previous work has established the philosophical foundations and demonstrated applications, here we articulate a comprehensive framework for analyzing and designing AI-enhanced education that honors complexity while maintaining normative orientation toward human flourishing.

A word about provenance is warranted at the outset. We develop multiplexity here using vocabulary drawn from the Islamic intellectual tradition, but the framework is pluralistic by design rather than parochial. Multiplexity is a salient feature of almost all classical philosophies and religions, from the ancient Greeks to the Abrahamic and Eastern traditions, prior to the rise of uniplex, reductionist approaches in the modern period; the layered, both-and habit of mind it expresses recurs in Heraclitean dialectic and Hegelian synthesis, in the Chinese *yin-yang*, in the Buddhist Middle Way, and in modern systems theory and complexity science. One can even observe, at present, a *multiplex turn* among thinkers who, aware of the limitations of either-or thinking, have grown discontent with it. What we highlight in this chapter are the Islamic roots

of this sensibility, with no claim to exclusivity; our aim is to contribute to this rising current by drawing on the Islamic intellectual tradition. Readers working from other traditions should find their own resources for layered thinking honored rather than displaced, and are invited to translate the framework into their own idioms.

1.1.3 Chapter Overview and Contributions

This chapter develops a comprehensive philosophical framework for understanding and ethically guiding the use of AI in educational contexts. Our core argument is that educational technology must be evaluated not merely by instrumental effectiveness but by its contribution to *kamāl*—the realization of human excellence and wholeness at individual and collective levels. We make four primary contributions.

First, we articulate *multiplex systemics* as a novel integration of systems theory and complexity science, Islamic philosophical concepts of unity (*tawḥīd*), perfection (*kamāl*), and self-development (*khudi*), dialectical philosophy and polarity management, and critical perspectives on technology and education. This synthesis offers a philosophically grounded yet practically applicable approach to educational technology design and evaluation.

Second, we challenge the implicit materialism and reductionism of much educational technology discourse by articulating a *stratified ontology* recognizing multiple, irreducible levels of reality—from the physical to the metaphysical—each with its own logic and ethical implications. This ontological reframing reveals dimensions of educational experience that technocratic approaches systematically overlook.

Third, we demonstrate how *polarity mapping*—a both-and approach to managing complementary opposites—can guide the design and implementation of educational AI in ways that honor competing values rather than forcing false choices. This methodological innovation provides practical tools for navigating the genuine tensions inherent in educational technology without collapsing into relativism.

Fourth, we ground our framework in a substantive account of human flourishing that goes beyond liberal neutrality to articulate positive ideals: the cultivation of wisdom (*ḥikma*), the development of integrated selfhood, and the advancement of just and spiritually grounded societies. This normative vision provides direction for technological development beyond mere efficiency or user satisfaction.

The remainder of this chapter proceeds as follows. Section 1.2 establishes theoretical foundations by tracing the evolution from either-or to both-and thinking and introducing key concepts of multiplexity and systemics. Section 1.3 presents the multiplex systemics framework in detail, including its philosophical grounding and practical principles. Section 1.4 applies this framework to educational technology, with extended analysis of AI writing assistants as a paradigmatic case. Section 1.5 explores broader implications for AI ethics in education and concludes with reflections on wholeness-making in an AI-mediated educational future.

1.2 Theoretical Foundations: From Either-Or to Both-And

1.2.1 The Dominance of Binary Logic and Its Limitations

To understand the need for multiplex systemics, we must first appreciate how deeply either-or thinking is embedded in modern intellectual culture. The roots of binary logic lie in classical Greek philosophy, particularly Aristotle's formulation of the law of excluded middle: a proposition must be either true or false, with no middle ground. This principle became foundational to Western logic, mathematics, and scientific method, enabling clear categorization, precise measurement, reproducible experiments, and ultimately digital computation itself.

The power of either-or thinking is undeniable. Yet as Korzybski (1933) warned nearly a century ago, this paradigm suffers from systematic fallacies when applied to complex realities: the fallacy of the single cause (attributing outcomes to one factor while ignoring multiple contributors), the fallacy of the excluded alternative (assuming that if one factor contributes, others do not), and the fallacy of misplaced concreteness (treating abstractions as concrete realities). These fallacies prove particularly problematic in educational contexts, where we might ask whether a student's struggles stem from individual ability, teacher quality, or curriculum design—when reality involves all these factors and many others interacting dynamically.

The oscillation between technological optimism and skepticism in educational technology reflects this binary thinking. As Wegerif and Major (2023) observe, discourse on educational technology tends to swing between announcing "the next best thing that will transform education" and condemning "the negative impact of technology on proper education." This pattern echoes ancient debates, as dramatized in Plato's *Phaedrus*, where Theuth praises writing as "an elixir of memory and wisdom" while King Thamus warns it will "produce forgetfulness" and create people with "the show of wisdom without the reality." Yet as Postman (1993) notes, "every technology is both a burden and a blessing; not either-or, but this-and-that." The challenge is moving beyond oscillation between extremes toward more nuanced understanding.

1.2.2 The Emergence of Both-And Thinking

Recognition of binary logic's limitations has prompted recurring calls throughout intellectual history for more integrative approaches. Dialectical philosophy, from Heraclitus's claim that "the way up and the way down are one and the same" (Graham, 2021) to Hegel's thesis–antithesis–synthesis framework (Hegel, 1977), holds that apparent opposites often express a deeper unity. For Hegel, contradiction is not error but the engine of development, suggesting that seemingly opposed pedagogical approaches—direct instruction versus discovery learning, individual versus collaborative work—need not be mutually exclusive but can generate higher-order practices integrating both.

Eastern philosophical traditions have long embraced complementary opposites. The Chinese concept of *yin-yang* embodies two forces simultaneously opposed and interdependent, each containing the seed of the other Siu (1957). Buddhist philosophy articulates the Middle Way (neither

asceticism nor indulgence) as not lukewarm compromise but a transcendent path honoring partial truths in opposing views while avoiding extremes. These wisdom traditions, as Song (2021) demonstrates in dialogue between AI and Chinese philosophers, offer resources for thinking beyond narrow "intelligence" paradigms toward more integrative understanding. Such traditions remind us that binary oppositions often reflect limitations in our conceptual frameworks rather than reality's structure—a recognition that proves essential for navigating AI's role in education.

Modern systems theory provides a scientific foundation for both-and thinking. As Capra (1996) demonstrates, living systems cannot be understood through reductionism alone. While analysis (breaking wholes into parts) remains necessary, it must be complemented by synthesis (understanding how parts form wholes). Key insights include emergence (wholes exhibiting properties unpredictable from parts alone), circular causality (effects feeding back to become causes), holons (entities simultaneously whole and part of larger wholes), and nested hierarchies (reality organized in levels, each with distinct principles).

McGilchrist (2009, 2021) offers a neuroscientific perspective grounded in extensive philosophical and empirical analysis, arguing that the brain's two hemispheres engage the world fundamentally differently: the left hemisphere focuses narrowly on details and categories (either-or), while the right hemisphere perceives broad patterns and relationships (both-and). Our civilization has become pathologically left-brained, privileging analysis over synthesis. In his comprehensive two-volume work, McGilchrist demonstrates how this hemispheric imbalance has shaped Western thought, leading to reductionism, mechanization, and loss of meaning—precisely the pathologies our multiplex systemics framework addresses. Wholesome education requires both perspectives in proper relationship, with the right hemisphere's integrative wisdom guiding the left hemisphere's analytical power.

Recent organizational theory has operationalized both-and thinking through *polarity management* (Johnson, 1992). Johnson distinguishes between problems to solve (situations with right and wrong answers resolvable once and for all) and polarities to manage (ongoing tensions between interdependent opposites requiring continuous balancing). Many challenges in education and technology are polarities, not problems: individual versus collaborative learning, structure versus flexibility, efficiency versus depth, innovation versus tradition, autonomy versus guidance. Treating polarities as problems—trying to "solve" them by choosing one side—inevitably backfires. Over-focusing on individual learning creates isolation; while an over-focus on collaboration undermines independent thinking. The solution is not picking one but leveraging both through oscillation between poles or designing systems honoring both simultaneously. Smith and Lewis (2022) document how leaders embracing both-and thinking outperform those stuck in either-or mindsets, finding creative syntheses ("the genius of AND") rather than forcing false choices ("the tyranny of OR").

1.2.3 Multiplexity: Reality as Layered and Integrated

While both-and thinking represents progress beyond binary logic, it remains insufficient without deeper understanding of how reality itself is structured. This is where the concept of *multiplexity* becomes crucial.

Şentürk (2012) articulates multiplexity as a core feature of Islamic intellectual tradition: "*A multiplex worldview accepts that existence, knowledge, values and truth have multiple levels and cannot be reduced to a single layer.*" This avoids both the extremes of *reductionism* (forcing everything into one level) and *relativism* (denying overarching unity or truth). Instead, multiple perspectives are held in a structured way: at some levels, truth can be relative (valid within context), while at another level, an ultimate Truth encompasses and transcends particulars. This formulation proves crucial for our purposes—multiplexity is not mere pluralism claiming "everything is relative," but rather recognizes *unity in multiplicity*: a hierarchical ordering where higher levels transcend and integrate lower levels, even as lower levels remain irreducible to higher ones.

Multiplexity derives from Latin *multiplex* (from *multus*, "many" and *plicare*, "to fold")—literally meaning "manifold" or "having many folds." Philosophically, multiplexity refers to reality as simultaneously stratified (organized in distinct levels), integrated (unified despite internal diversity), complementary (containing apparent opposites that actually support each other), and contextual (exhibiting different truths at different levels or perspectives).

A multiplex worldview entails what Hartmann (1952) calls *stratified ontology*—reality structured in ontological levels, each with organizing principles. Consider the physical level (matter-energy governed by physical laws), biological level (living organisms with emergence of life, metabolism, reproduction), psychological level (consciousness, intentionality, subjective experience), social level (collective practices, institutions, cultures), and spiritual level (transcendent meaning, ultimate purpose, connection to the divine). Each level exhibits properties that cannot be reduced to lower levels, yet depends on lower levels as a necessary foundation. Life requires physics, but cannot be fully explained by physics alone. Mind emerges from biology but has causal powers not reducible to neural firings. Social realities like "democracy" or "education" are real and causally efficacious despite not being physical objects.

This has profound implications for educational technology. If humans are stratified beings—simultaneously physical, biological, psychological, social, and spiritual—then educational interventions must attend to all these dimensions. An AI system optimized solely for cognitive efficiency (psychological level) while ignoring social belonging or spiritual meaning will inevitably produce partial and potentially harmful outcomes.

Islamic philosophy explicitly theorizes this stratification through the concept of *marātib* (levels or stages), with classical scholars identifying multiple levels of existence (*marātib al-wujūd*): *mulk* (the visible, material world), *malakūt* (the invisible, angelic realm), *jabarūt* (the realm of divine power and archetypes), and *lāhūt* (the divine essence itself). Paralleling these are levels of knowledge (*marātib al-ʿulūm*) and levels of methods (*marātib al-uṣūl*). Classical Muslim scholars moved

fluidly between disciplines (law, theology, philosophy, mysticism) because they understood these as different lenses on unified reality (Şentürk et al., 2020).

1.2.4 Systemics: Interconnection, Feedback, and Emergence

If multiplexity addresses the *vertical* dimension (levels of reality), systemics addresses the *horizontal* dimension (relationships within and across levels). The term *systemics* itself offers instructive contrast with "science."

Cybernetician Heinz von Foerster provided illuminating etymology: he observed that "science" derives from the Indo-European root *scy*, meaning "to separate"—the same root found in "schizophrenia" and "schism." Science has been remarkably successful, he noted, precisely through this process of separation and analysis. Yet "systemics" represents the exact opposite: integration. As von Foerster argued:

> When you think about it today, all this systems theory and systems research which crops up in both art and science, I wouldn't call that 'science' anymore, I would call that 'systemics.' Today's science has moved on to an approach that sees things together: systemics. So, I would see the steps taken today as being from science to systemics. (von Foerster, 2003)

This shift from science to systemics parallels the movement from either-or to both-and thinking, from analysis to synthesis, from abstract universal principles to contextual integration. Systems thinking shifts focus from isolated objects to networks of relationships, from linear causality to circular feedback, from static states to dynamic processes (Meadows, 2008). Core principles include interconnection over isolation (no entity exists independently; everything embeds in webs of relationships partly constituting what it is), circular causality and feedback loops (A affects B, which affects C, which feeds back to affect A, creating reinforcing or balancing dynamics), emergence and downward causation (collective behavior exhibiting properties unpredictable from individual components, with emergent wholes exerting causal influence on their parts), and leverage points for systemic intervention (small changes at key points producing large systemic effects, with the most powerful leverage involving paradigm shifts—changes in underlying assumptions and mental models) (Meadows, 1999).

Morin (2001) articulates "complex thinking" as essential for navigating contemporary challenges, proposing seven complex lessons crucial for education's future. His concept of *unitas multiplex* (unity in multiplicity) aligns precisely with our multiplex systemics framework. Morin warns against the "blind intelligence" produced by specialization that "abstracts" knowledge from context, breaking "the systemicity (relation of a part to the whole) and multidimensionality of phenomena." For Morin, education must teach methods of "*grasping mutual relations and reciprocal influences between parts and the whole in a complex world.*" He insists education illustrates the human unity-and-diversity: "*Education of the future should be careful to not allow the idea of the unity of the*

human species to efface the fact of its diversity, or the idea of its diversity to efface its unity." This both-and principle—honoring unity *and* diversity simultaneously—epitomizes multiplex thinking.

Morin distinguishes between problems (which may be solved through application of rigorous method) and what Schumacher (1977) calls "divergent problems" (which require wisdom and cannot have final solutions). Education represents a divergent problem par excellence—there is no single solution, it is inherently dynamic and complex, it requires continuous balancing of opposites, it cannot be solved once and for all but demands that every society deal with it afresh. Treating education as merely a technical problem to be "solved" through better algorithms or optimized systems represents profound category error.

1.2.5 Wegerif's Dialogic Theory and Educational Technology

Wegerif and Major (2023) argues compellingly that educational theory must take the voice of technology seriously rather than treating it as neutral tool. Drawing on Heidegger's phenomenology, Wegerif observes that technology is not merely instrumental but shapes us from inside, influencing how we understand ourselves and make decisions. This resonates deeply with our multiplex systemics approach—technology operates simultaneously at multiple levels (technical, pedagogical, psychological, social, political, spiritual), with effects rippling across these levels in complex feedback loops.

Wegerif identifies three "grammars" or ways of framing educational technology: as teaching machine (technology delivering optimized instruction), as tool to think with (technology augmenting cognitive capabilities), and as learning environment (technology mediating social learning). Each grammar embodies different assumptions about agency, learning, and human nature. The teaching machine grammar, exemplified by Skinner's behaviorist approach and contemporary adaptive learning systems, positions technology as agent replacing or directing human teaching. The tool-to-think-with grammar, exemplified by Papert's constructionism and LOGO programming, emphasizes learner agency and cognitive empowerment. The learning environment grammar, exemplified by Bereiter and Scardamalia's Knowledge Forum, focuses on technology mediating collaborative knowledge building.

From our multiplex systemics perspective, these grammars need not be mutually exclusive. Rather than choosing one "correct" grammar, we can recognize each as highlighting different dimensions of educational technology that must be held in productive tension. A comprehensive approach to AI in education requires elements of all three: efficient transmission of established knowledge (teaching machine), development of individual capabilities for thought and creation (tool to think with), and induction into communities of practice and collective intelligence (learning environment). The art lies in balancing these complementary aspects appropriately for context, learner, and educational goals—a both-and integration rather than either-or choice.

Wegerif's dialogic theory proves particularly valuable because dialogue itself operates multiplexly—at the epistemological level as means of knowledge construction, at the ontological level as constitutive of selfhood and reality, and at the pedagogical level as method for drawing learners into

educational participation. As Wegerif argues, "learning requires dialogue and so is first called into being by someone or something outside the self." The dialogic relation makes learning possible precisely through the learner being "partly outside themselves entangled up with the other." This challenges atomistic individualism underlying much educational technology, suggesting that AI systems treating learners as isolated units fundamentally misunderstand the relational nature of human development and learning.

1.2.6 Toward Multiplex Systemics: Integrating Levels and Relations

Multiplex systemics synthesizes stratified ontology (multiple levels) with systems thinking (dynamic relations). It recognizes that reality is organized in levels (physical, biological, psychological, social, spiritual), each level has its own principles and emergent properties, levels are interconnected through bottom-up emergence and top-down causation, within each level entities are embedded in networks of feedback and interdependence, understanding requires multiple perspectives and methodologies appropriate to each level, and ethical action requires attending to all levels and their interactions.

Taken together, these commitments make multiplex thinking a stance of *openness* that can be read along three structural dimensions (Şentürk, 2012), shown in Figure 1.1: *horizontal* openness (openness to perspectives: holding apparent opposites in productive tension through both-and thinking and polarity navigation), *vertical* openness (openness to transcendence: ordering values so that higher purposes guide lower means rather than the reverse), and *dynamic* openness (openness to emergence: treating education as a living system that unfolds in time, marked by feedback, change, and irreducible particularity). We prefer *dynamic* to *systemic* here to foreground temporal as well as structural complexity. This openness is also constitutively plural in two further senses: it is *open-civilizational* (hospitable to many traditions, as our earlier remarks on provenance stressed) and *open-capability* (hospitable to many learners, refusing a single fixed template of the human against which difference reads as deficit). The same disposition can be described as a movement beyond reductionism along four facets, each countering a distinct reductionist error on a distinct axis (Figure 1.2): openness to *wholeness* (against fragmentation, mistaking the part for the whole), *possibility* (against closure, mistaking one frame for the truth), *aliveness* (against mechanization, mistaking the living for the machine), and *integration* (against reification, mistaking the representation for reality). Ellen Langer's *mindful learning*, with its insistence on teaching conditionally and viewing the same material from several perspectives at once, names the pedagogical correlate of such openness (Langer, 1997).

The two figures describe one disposition from two angles, not two rival taxonomies. The three dimensions name *where* openness operates—horizontal, vertical, and dynamic; the four facets name *what* reductionism each undoes. They line up naturally: openness to wholeness and to integration belong with the systemic-holistic work of seeing parts within wholes; openness to possibility is the horizontal, both-and refusal of premature closure; and openness to aliveness is the dynamic attention to what changes and emerges. Integration is the meta-discipline that holds the others

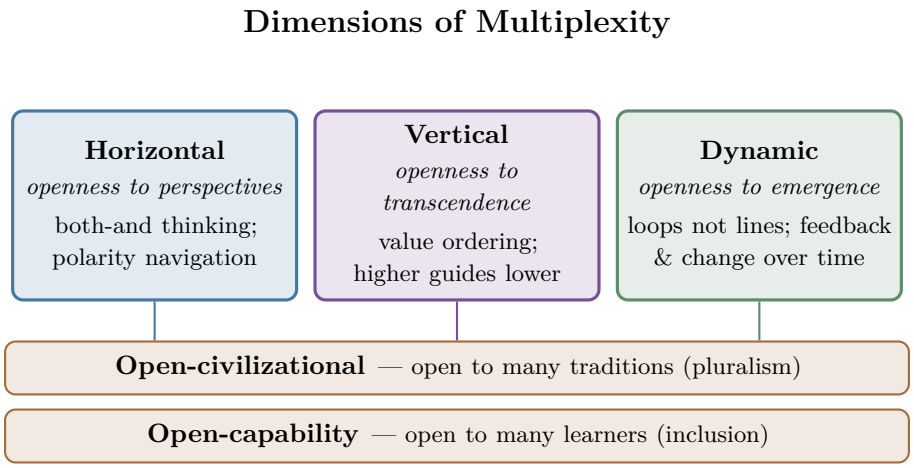


Figure 1.1: **Dimensions of multiplexity.** Horizontal (openness to perspectives), vertical (openness to transcendence), and dynamic (openness to emergence) openness, cross-cut by two pluralizing commitments: *open-civilizational* (hospitable to many intellectual traditions) and *open-capability* (hospitable to many learners, including neurodivergent and disabled students). Together they position multiplexity as an invitation rather than an enclosure.

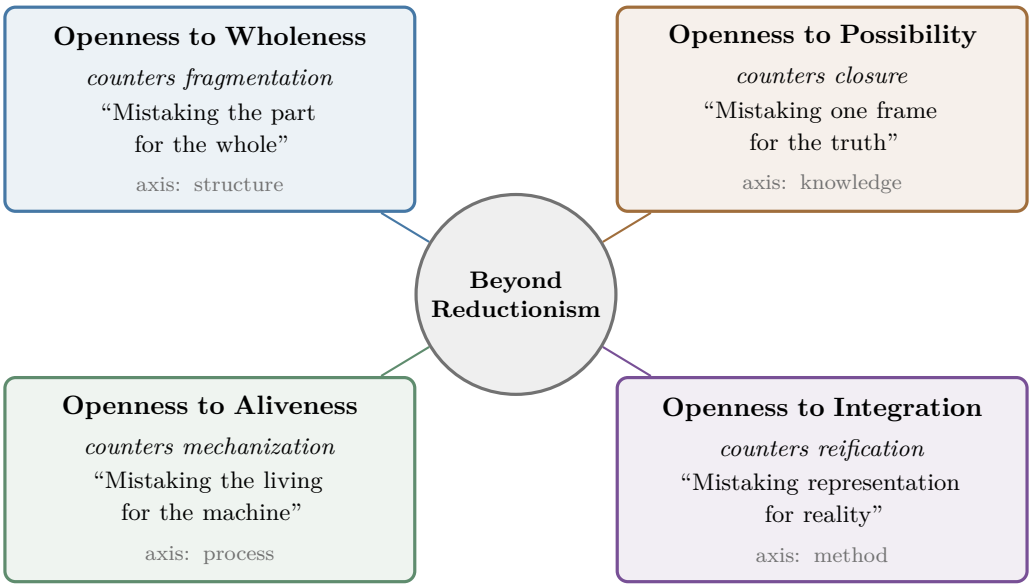


Figure 1.2: **Four kinds of openness in multiplexity.** Each facet counters a distinct reductionist error on a distinct axis, so the four are mutually exclusive and jointly exhaustive: *wholeness* against fragmentation (structure), *possibility* against closure (knowledge), *aliveness* against mechanization (process), and *integration* against reification (method)—using reduction as a tool without mistaking the representation for reality.

together, using reduction without absolutizing it, which is itself a vertical, value-ordering act that subordinates method to the whole it serves.

This openness has a cognitive correlate in Langer's contrast between *intelligence* and *mindfulness*: intelligence seeks the optimal fit between learner and environment and moves directly from

problem to solution, whereas mindfulness holds several perspectives at once, sees the familiar as novel, and treats meaning as conferred on outcomes rather than given in advance (Langer, 1997). Mindfulness is thus multiplexity in the register of individual cognition, though Langer's contrast captures only its horizontal and dynamic dimensions and is silent on the vertical ordering toward *kamāl* and the transcendent that our *tawḥīdic* framing supplies.

Applied to educational technology, multiplex systemics insists that AI systems cannot be evaluated solely on technical metrics (efficiency, accuracy) or even first-order ethical criteria (fairness, privacy). We must ask:

- *ontologically* (what conception of the human person is embedded in this technology?)
- *epistemologically* (what forms of knowledge are privileged or marginalized?)
- *methodologically* (how do we assess impacts across multiple levels and timescales?)
- *ethically* (does this technology move us toward wholeness or fragmentation?), and
- *teleologically* (what ultimate purposes and ideals does it serve?).

These questions set the stage for the normative framework we develop in the next section: wholeness-making as a guiding ideal for educational technology in a multiplex systemics paradigm.

1.3 The Multiplex Systemics Framework

1.3.1 Wholeness-Making as Normative Ideal: The Concept of Kamāl

At the heart of our multiplex systemics framework lies a normative vision: the ideal of *kamāl*. This Arabic term, often translated as "perfection" or "excellence," connotes completion, wholeness, and fullness of being (Nasr, 2007). To understand *kamāl* is to grasp that perfection is not about flawless uniformity but about realization of potential: the actualization of latent capacities toward their highest expression. This ideal connects closely to the concept of *iḥsān* (excellence), the highest level of practice where one acts with full consciousness and beauty, as if in the divine presence (Qadir et al., 2020). The opposite of *kamāl* is *naqs* (deficiency/incompleteness), understood not simply as having flaws but as *fragmentation*—failure to integrate and harmonize one's diverse capacities. The four reductionisms that Figure 1.2 sets against openness are not merely epistemic errors; each is a mode of *naqs*, the fragmentation that is the privation of *kamāl*. The four facets of openness are therefore the movements of wholeness-making: openness to wholeness, possibility, aliveness, and integration are how education repairs, respectively, the fragmented, the foreclosed, the mechanized, and the reified self, drawing its diverse capacities back toward *kamāl*.

Islamic philosophy describes humans as possessing multiple faculties that must work together harmoniously (al Ghazali, 2010). The intellect (*'aql*) gives us the capacity for reason and knowledge. The heart (*qalb*) is the seat of emotions and spiritual intuition. The soul or self (*nafs*) acts as the

integrating center of will and identity, bringing all faculties together. The body (*jism*) is our physical vessel with its needs and drives. *Kamāl* requires that all these dimensions function in harmony: reason guiding desires, the heart attuned to higher values, and the body disciplined to serve rather than dominate. *Naqs* arises when faculties become disconnected—intellect divorced from ethics, emotions untempered by wisdom, bodily appetites overwhelming higher purposes.

The concept of *khudi*, introduced by philosopher-poet Muhammad Iqbal (1877–1938), describes the dynamic process of wholeness-making. *Khudi* refers to the self or ego understood not as a fixed essence but as an ongoing project of self-construction and self-transcendence (Iqbal, 1930). Iqbal conceived *khudi* as the principle of individuality and creative agency—the self as active organizer rather than passive recipient. A powerful metaphor comes from Iqbal's interpreter Rafiuddin, who compares the self to a coach pulled by many horses (representing different desires and impulses). Without a driver, the coach moves fitfully, pulled in conflicting directions by whichever horse is momentarily strongest. With a strong driver (developed *khudi*), the person marshals diverse energies toward coherent ends. This metaphor proves crucial for educational ethics: education, rightly understood, cultivates this "inner driver"—developing capacity for self-direction, delayed gratification, and value-guided action. Technologies that bypass or atrophy this capacity (through addictive design, algorithmic micro-management, or passive content delivery) produce naqs even if they yield short-term performance gains.

The Islamic principle of *tawḥīd* (divine unity) provides a metaphysical foundation for wholeness-making. *Tawḥīd* is not merely monotheism but a comprehensive worldview seeing reality as an integrated whole. It implies ontologically that all existence is interconnected, epistemologically that all true knowledge forms a coherent whole, and ethically that all authentic values ultimately harmonize. This enables what Şentürk (2012) calls "unity in multiplexity": acknowledging many truths and realities without asserting all are equal. There is an ultimate integrative framework (divine unity) that does not flatten differences but orchestrates them. Applied to education, *tawḥīd* suggests that knowledge should not be fragmented into isolated disciplines but taught as an interconnected whole, that intellectual-moral-spiritual development should be integrated rather than separated, that individual excellence and collective welfare are mutually supporting rather than opposed, and that education serves the ultimate purposes of human existence rather than merely economic or political ends.

A brief comparison with other wholeness traditions illuminates the universality of this ideal. Heidegger's concept of *poiesis* (creative bringing-forth that reveals hidden aspects of reality) and his call for education fostering questioning of Being resonate with the Islamic concept of *taḥqīq* (realization of truth in one's being, not just conceptually but existentially). Both traditions emphasize that authentic knowledge involves the transformation of self, not merely the accumulation of information. This convergence between Western phenomenological philosophy and Islamic pedagogy suggests that wholeness-making represents a perennial human ideal rather than a culturally parochial notion, recognized across diverse intellectual traditions that attend to the full dimensions of human development.

From this foundation, we can articulate *wholeness-making* as a guiding ideal for educational technology ethics. Wholeness-making is the process of integration (harmonizing diverse capacities into a coherent self), actualization (realizing latent potentials toward highest expression), contextualization (situating individual development within larger wholes), and orientation (directing growth toward transcendent ideals beyond mere adaptation). A technology supports wholeness-making to the extent that it honors stratified ontology by attending to all dimensions of the person, strengthens agency by developing rather than bypassing capacities for self-direction, fosters integration by connecting rather than fragmenting aspects of learning, serves transcendent purposes by aiming beyond mere efficiency, supports relational embedding by enhancing rather than substituting human relationships, and enables contextual wisdom by cultivating judgment appropriate to situations. Figure 1.3 draws the framework together: the stratified levels of reality, linked by emergence and downward causation and bound within each level by dynamic feedback, oriented as a whole toward *kamāl*.

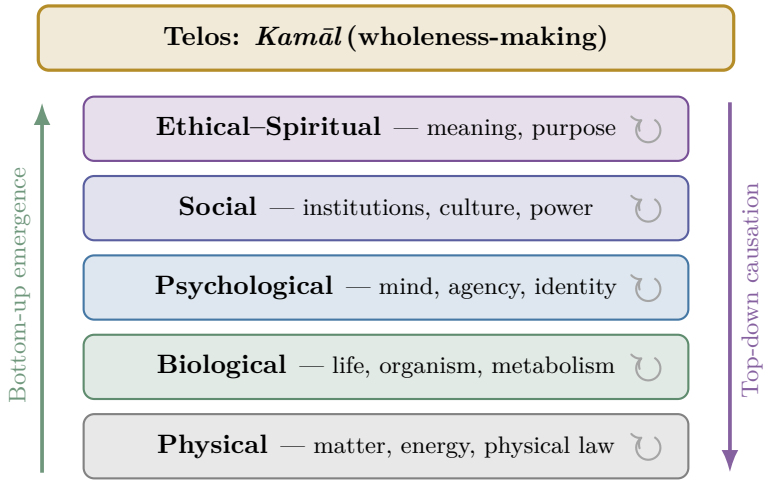


Figure 1.3: **The multiplex systemics framework.** Stratified levels of reality, each with irreducible properties, are connected vertically by emergence and downward causation and held together by dynamic feedback that unfolds over time. Educational technology is to be evaluated not at one level alone but across all of them, oriented toward *kamāl* (wholeness-making).

This orientation has direct purchase on inclusive education. Because multiplexity begins from many irreducible levels and many valid perspectives, it is inherently *open-capability*: it refuses any fixed, static template of a "standard learner" against which difference registers as deficit. Neurodivergent and disabled learners are then understood not as deviations to be remediated toward a norm but as distinct configurations of capacity whose flourishing the framework is obliged to attend to in its own terms—a stance congruent with Langer (1997), whose *mindful learning* teaches conditionally and holds several perspectives on the same material at once, so that learners may engage it through the route that fits them. Polarity management, too, is a practical resource for inclusion: tensions such as structure and flexibility, standardization and personalization, or

independence and support are precisely the dimensions along which accessible and neuro-affirming designs must continuously rebalance for different learners, rather than resolving them once for an imagined average.

1.3.2 Core Principles of Multiplex Systemics

Having established wholeness-making as a normative ideal, we now articulate core principles operationalizing multiplex systemics for educational technology analysis and design.

Principle 1: Recognize Multiple Levels (Marātib). Educational phenomena must be understood at appropriate levels of analysis without illegitimate reduction or category confusion. The key levels for educational technology are the technical level (algorithms, interfaces, data structures), pedagogical level (teaching methods, learning activities, assessment practices), psychological level (motivation, cognition, emotion, identity), social level (classroom dynamics, institutional practices, power relations), political-economic level (resource allocation, policy frameworks, market forces), and ethical-spiritual level (values, meanings, purposes, ultimate concerns). Each level has irreducible properties and appropriate methods of inquiry. Technical analysis cannot answer pedagogical questions; psychological research cannot resolve ethical dilemmas; economic metrics cannot capture spiritual meaning. However, levels are interconnected—technical design shapes pedagogical possibilities, psychological dynamics influence social patterns, political-economic contexts constrain ethical choices. Multiplex analysis attends to both the integrity of each level and interactions between levels.

Principle 2: Identify and Manage Polarities. Educational technology involves numerous tensions between complementary values. Following Johnson (1992), we distinguish problems to solve (situations with right and wrong answers resolvable once and for all) from polarities to manage (ongoing tensions between interdependent opposites requiring continuous balancing). Key polarities in educational AI include efficiency and depth, personalization and standardization, automation and human judgment, innovation and tradition, autonomy and guidance, individual work and collaborative learning, assessment and learning, skills and knowledge. The "tyranny of OR" treats these as forced choices; the "genius of AND" seeks to leverage both poles (Johnson, 2020a,b). Managing polarities requires mapping upsides and downsides of both poles, recognizing that over-focus on one pole eventually triggers its downsides, designing systems that oscillate between poles or integrate both simultaneously, and monitoring indicators to detect when adjustment is needed.

Principle 3: Attend to Feedback Loops and System Dynamics. Educational interventions trigger cascading effects through feedback loops. Rather than assuming linear causality, multiplex systemics maps these dynamics. For example, introducing an AI tutoring system may initiate a virtuous cycle (personalized support leads to success, which increases motivation, which produces more engagement, yielding improved outcomes and greater confidence) or a vicious cycle (AI does cognitive work for students, leading to skill atrophy, increased dependence, reduced self-efficacy, avoidance of challenge, further decline, and learned helplessness). Which dynamic emerges depends

on design choices, implementation context, and system-level factors. Multiplex analysis anticipates these dynamics rather than assuming isolated effects.

Principle 4: Design for Both-And Integration. Rather than forcing either-or choices, multiplex design seeks both-and solutions through temporal oscillation (alternating between poles over time), spatial distribution (different spaces emphasizing different poles), hierarchical nesting (one pole serving another at higher level), and dialectical synthesis (creative integration transcending original opposition). The key is avoiding what Johnson calls the "tyranny of OR" (seeing poles as mutually exclusive) and the "tyranny of AND" (collapsing into relativism where everything is acceptable), instead achieving what we might call the "wisdom of OR *with* AND"—maintaining capacity for decisive judgment (sometimes one pole is appropriate) while remaining open to integration (often both poles are needed).

Principle 5: Orient Toward Kamāl (Wholeness-Making). Every design choice should be evaluated not merely by efficiency metrics but by contribution to wholeness-making. This requires asking whether technology strengthens or weakens human agency, integrates or fragments different dimensions of development, serves short-term performance or long-term flourishing, honors human dignity and transcendent purpose, and builds or erodes authentic human relationships. This principle provides normative orientation preventing multiplex systemics from collapsing into a value-neutral description of complexity.

1.3.3 Polarity Management as Practical Method

To operationalize multiplex systemics principles, we employ *polarity management*—a structured approach for navigating tensions between complementary values (Johnson, 1992). Unlike problems that can be solved once and for all, polarities are ongoing tensions between interdependent opposites, both of which are necessary. The goal is not to choose one pole over the other but to leverage the strengths of both while managing their inherent tensions.

Consider the fundamental educational polarity of self-reliance versus technology-reliance. Self-reliance offers deep learning through struggle, authentic voice, and intrinsic motivation, but risks slow inefficiency, limited perspective, and cognitive fatigue. Technology-reliance provides speed, personalized scaffolding, and immediate feedback, but risks shallow engagement, skill atrophy, and loss of agency. Effective practice requires both: meaningful learning developing the whole person through dynamic integration of human agency and technological support.

Managing polarities requires strategic action steps that leverage both poles: teach explicit human-AI collaboration strategies, design assignments requiring both human creativity and AI assistance, build metacognitive reflection on tool use, and assess both independent capabilities and collaborative effectiveness. It also requires vigilance—monitoring early warning indicators of over-focus on either pole and adjusting the balance when warnings appear by deliberately moving toward the neglected dimension.

A useful tool for analyzing polarities is *polarity mapping*, which visualizes the upsides and downsides of each pole alongside the higher purpose their integration serves. This approach proves

especially powerful for the most foundational educational tension: inner development versus outer competence.

1.3.4 Inside-Outside Multiplexity: *Tazkiyah* and *ta'lim*

The wholeness-making framework finds practical application in what we might call *inside-outside multiplexity*—the recognition that genuine education requires both inner moral development and outer technical competence. Drawing on Islamic pedagogical philosophy, Qadir and Senturk (2026b) articulate this as the complementary relationship between *tazkiyah* (moral purification and character formation) and *ta'lim* (knowledge transmission and skill development). This dual focus is rooted in Islamic practice, which emphasizes multiple dimensions of learning encompassing both outward practice and inward spiritual development (Ahmed, 2021).

This distinction becomes especially critical in the AI era. As knowledge becomes increasingly commodified, available instantly through search engines and language models, what distinguishes educated persons from information processors is not what they *know* but how they *act*: the ethical judgment they exercise, the responsibility they assume, the wisdom they embody. Technical AI literacy without moral formation risks producing "skilled incompetence": individuals capable of deploying powerful technologies but lacking ethical frameworks to use them wisely. Concepts such as *iḥsān* (spiritual excellence) and *futuwwah* (altruistic ethics) emphasize acting with sincerity, selflessness, and moral responsibility (Senturk, 2023; Qadir et al., 2020). This orientation toward communal care and ethical conduct contrasts sharply with the individualistic and competitive ethos often characterizing modern education and its technologies.

Conversely, moral aspiration without practical competence remains ineffectual. The contemporary world demands that educators cultivate *both* the inner compass (character, values, ethical sensibility, reflective capacity) *and* the outer capabilities (AI literacy, technical skills, information fluency, practical effectiveness). To visualize this polarity, we employ a polarity map (as shown in Figure 1.4).

A polarity map consists of four quadrants arranged around two poles. Each pole has upsides (marked with +) representing what we gain when we do that pole well, and downsides (marked with –) representing what we lose when we over-focus on that pole to the neglect of the other. Above the poles sits the Greater Purpose Statement (GPS)—the aspirational outcome we achieve by leveraging both poles synergistically. Below sits the Greatest Fear—the negative outcome when we experience only the downsides of both poles. The infinity loop pattern illustrates that managing polarities is not a one-time choice but requires ongoing dynamic movement between poles.

This inside-outside multiplexity aligns with our earlier criteria for wholeness-making technologies. Educational AI should be designed to strengthen *both* poles: developing students' moral reasoning and ethical discernment (inner) while simultaneously building their technical capabilities and information fluency (outer). Systems that enhance one dimension while atrophying the other fail the wholeness test.

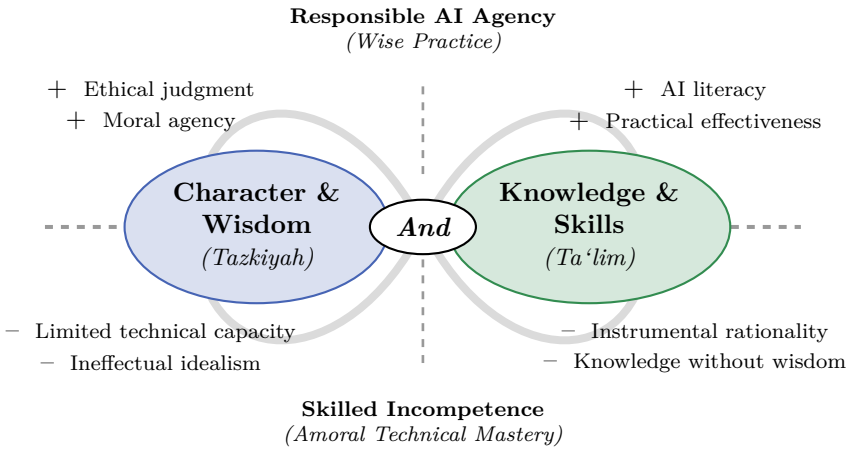


Figure 1.4: **Polarity Map: Inside-Outside Multiplexity**. Character development (*tazkiyah*) and technical competence (*ta'lim*) form complementary poles, each with predictable upsides and downsides. The Greater Purpose—*Responsible AI Agency*—emerges when both poles are leveraged synergistically. The Greatest Fear—*Skilled Incompetence*—manifests when we experience only downsides. Effective education requires ongoing dynamic movement between both poles, as Qadir and Senturk (2026b) emphasize.

1.4 Applying Multiplex Systemics to AI in Education

1.4.1 Extended Case Analysis: AI Writing Assistants in Higher Education

To demonstrate the power of multiplex systemics, we provide an extended analysis of AI writing assistants—tools like ChatGPT, Claude, and others that can generate, revise, and critique written work. This case proves particularly instructive because it involves tensions educators currently grapple with and reveals multiple levels of analysis essential for ethical decision-making.

Technical Level Analysis

At the technical level, large language models (LLMs) are trained on massive text corpora and generate coherent, contextually appropriate text through pattern matching and statistical generation. Key features include:

1. *Fluency without understanding*—LLMs produce grammatically correct text that may lack factual accuracy or logical coherence.
2. *Variable capability*—performance depends on training data quality and model architecture.
3. *Absence of consciousness*—no internal beliefs, intentions, or genuine comprehension.

Ethical considerations at this level include:

1. *Bias in training data*—models reproduce stereotypes and prejudices present in source material.
2. *Environmental impact*—energy consumption of training and running large models raises sustainability concerns.

3. *Concentration of power*—a few corporations control the most advanced systems, creating dependencies.

While these technical considerations matter, a multiplex approach insists they represent only one level of analysis. Understanding AI writing assistants solely as technical artifacts profoundly misses their significance.

Pedagogical Level Analysis

At the pedagogical level, writing instruction aims to develop multiple interconnected capacities: critical thinking and argumentation, evidence synthesis and evaluation, rhetorical awareness and audience sensitivity, revision practices and metacognitive monitoring, and knowledge construction through the act of writing itself. Key pedagogical questions include:

- *What is the purpose of writing assignments*? (Demonstrating knowledge? Developing thinking? Communication practice?)
- *What role could AI play in this process*? (Generating ideas? Providing feedback? Automating drafting?)

The pedagogical challenge is that writing development involves not just producing text but transforming thought. As writing scholars have long argued, writing is not merely a transcription of pre-formed ideas but a generative process through which thinking develops (Flower and Hayes, 1981). If AI short-circuits this process by generating text directly, students may produce better-looking products while developing less robust thinking capabilities. However, if AI is positioned as collaborator in the thinking process—helping generate ideas to critique, providing quick feedback enabling more revision cycles, or modeling expert writing for analysis—it might enhance rather than undermine development.

Ethical considerations at the pedagogical level include risk of short-circuiting developmental processes essential for skill formation, opportunity for enhanced feedback and scaffolding enabling more effective learning, and the need to rethink assignment design to focus on dimensions AI cannot automate (original analysis, personal synthesis, authentic voice). Wegerif and Major (2023) observes that educational technology shaped historically by "teaching machine" logic focuses on efficient content delivery; the challenge with AI writing assistants is preventing reduction of writing to mechanical production while preserving the dialogic dimension—writing as thinking, as a relationship with an imagined reader, as participation in scholarly community.

Psychological Level Analysis

At the psychological level, writing development involves both cognitive skills (planning, organizing, revising, evaluating) and identity formation (finding one's voice, developing authorial confidence, seeing oneself as a capable writer). Key psychological dynamics include:

1. *Skill formation through struggle*—productive failure and difficulty build capabilities that persist beyond the immediate task.
2. *Skill atrophy through dependence*—overreliance on external tools can weaken abilities through systematic disuse.
3. *Identity as writer*—authorial selfhood requires experience of authentic production, not merely editing AI-generated text.
4. *Motivation and agency*—sense of authorship and genuine accomplishment affects intrinsic drive to engage and improve.

Research in self-determination theory shows that human motivation depends on experiencing *autonomy* (sense of self-direction), *competence* (sense of capability), and *relatedness* (sense of meaningful connection) (Ryan and Deci, 2000). AI writing assistants potentially threaten all three: autonomy is diminished if AI directs the writing rather than serving the writer's intentions, competence is undermined if the writer doubts whether accomplishments are truly their own, and relatedness may be weakened if writing becomes a transaction with the machine rather than communication with a human audience.

However, thoughtfully designed AI could support these needs: providing scaffolding that enables accomplishment (competence), offering responsive feedback (relatedness), while remaining under writer's control (autonomy). The concept of khudi proves particularly relevant here. Developing writing capability means developing the "inner driver"—executive function that sets goals, monitors progress, and persists through difficulty. AI positioned as a tool the writer consciously deploys may strengthen khudi; AI positioned as an authority to defer to, or content producer the writer merely edits, risks weakening it.

This distinction between strengthening and weakening agency surfaces key ethical considerations at the psychological level: Does AI dependence erode self-efficacy and writer identity? Does it support or substitute for executive function and metacognitive development? Are we providing scaffolding (temporary support enabling eventual independence) or bypassing (permanent substitution preventing skill formation)? These questions prove crucial because the answers determine whether AI serves wholeness-making or fragmentation.

Social Level Analysis

At the social level, academic writing occurs within communities with norms about authorship, originality, collaboration, and intellectual integrity. Key social tensions include:

1. *Contested definitions of authorship and plagiarism*—is AI-assisted text fundamentally different from using grammar checkers?
2. *Shifting cultural norms*—will widespread AI use raise or lower standards for student writing?
3. *Differential access to tools*—will AI widen or narrow achievement gaps?
4. *Evolving power dynamics*—who decides how much transparency is required and what counts as acceptable assistance?

Academic culture rests on implicit agreements about what writing represents. When reading a student essay, instructors traditionally assumed it reflected that student's thinking, developed through their effort, expressed in their voice. AI writing assistants disrupt these assumptions. As Sparrow and Flenady (2025) provocatively argue, automated education risks producing "bullshit universities" where neither teachers nor students engage authentically—teachers assign essays generated by AI, students submit essays generated by AI, and everyone goes through motions of education without genuine learning occurring. While this dystopian vision is deliberately provocative, it highlights real social dangers of normalized inauthenticity.

However, social norms are not static. Academic culture has adapted before to new writing technologies—printing, typewriters, word processors, spell-checkers, online databases. Each raised concerns about authenticity and skill degradation, yet ultimately were integrated in ways that transformed expectations without destroying education's core purposes. The challenge is managing this transition thoughtfully rather than allowing it to happen by default.

Ethical considerations at the social level include the need for community discourse establishing new norms rather than having norms imposed unilaterally, potential for widening or narrowing achievement gaps depending on who has access to the best tools and preparation for using them effectively, impacts on academic culture of trust and intellectual integrity, and power dynamics around surveillance and control versus transparency and shared governance.

Political-Economic Level Analysis

At the political-economic level, AI writing tools are commercial products embedded in larger techno-economic systems. Key political-economic factors include:

- *Unequal access based on financial resources*—do premium AI tools create a two-tiered educational system?
- *Dependency on proprietary platforms*—does writing capability become contingent on corporate access?
- *Data harvesting and privacy concerns*—is student writing being fed into corporate databases without adequate protection?
- *Market pressures toward automation*—do economic incentives favor replacing human instruction with technological substitutes?

Birhane (2021) emphasizes that algorithmic systems are never neutral but reflect and reproduce existing power relations. The rapid deployment of AI in education is driven largely by corporate interests seeking profitable markets, not by pedagogical wisdom about what serves learning. Moreover, as with previous educational technologies, the benefits and harms are distributed unequally. Well-resourced institutions can integrate AI thoughtfully with strong human support; under-resourced institutions may use AI as a substitute for adequate human instruction, widening educational inequality.

Ethical considerations at the political-economic level include issues of *access and equity* (who benefits from AI writing tools, who is harmed?), *concentration of power* (should educational infrastructure depend on corporate platforms?), *implications for labor* (will writing instruction be devalued? will humans be displaced?), and *tension between public good and private profit.*

Ethical-Spiritual Level Analysis

At the ethical-spiritual level, writing relates to fundamental questions of human meaning, authenticity, and self-expression. Deep questions at this level include:

1. *Autonomy and selfhood*—what does it mean to think for oneself when AI can generate our thoughts?
2. *Formative struggle*—how important is creative effort and difficulty to human flourishing?
3. *Meaning-making*—what is lost when we outsource interpretation and expression to machines?
4. *Wisdom versus processing*—how do we cultivate judgment and not just information manipulation?
5. *Intrinsic value*—is there worth in wrestling with ideas and crafting expressions independent of product quality?

Many wisdom traditions emphasize the formative value of sustained effort and discipline. The Islamic concept of *mujāhada* (spiritual struggle) suggests that wrestling with difficulty—including the difficulty of expressing complex thoughts in writing—is not merely means to an end but intrinsically valuable for developing character and deepening understanding. Similarly, Heidegger's emphasis on questioning and dialogue as paths to authentic thought suggests that too-easy answers (such as those AI can generate instantly) may prevent the deeper thinking that emerges only through sustained grappling.

As Birhane (2024) argues, certain dimensions of education (particularly meaning-making, ethical judgment, and authentic connection) are fundamentally "incomputable," resisting reduction to algorithmic processes without losing what makes them educationally valuable. This incomputability reflects not a technical limitation but the ontological nature of human experience. Wrestling with ideas to forge authentic expression, developing voice through sustained effort, connecting meaningfully with human audiences—these spiritual and ethical dimensions of writing resist quantification and optimization.

Key ethical concerns at this level thus include:

- *Risks of alienation and inauthenticity*—when writing becomes performance for algorithms rather than genuine human expression
- *Loss of space for meaning-making*—reduction of writing from transformative practice to mere information transfer
- *Tension between efficiency and depth*—forms of knowing that require time, struggle, and contemplation resist optimization

The ethical-spiritual analysis, however, need not be negative. If AI writing assistants can be positioned as partners in dialogue rather than a replacement for thinking, they might enrich rather than impoverish the meaning-making process. The key distinction is between using AI as an oracle (source of ready-made answers) versus using AI as an interlocutor (partner in the thinking process). As Wegerif and Major (2023) observes, dialogue is not merely a means to knowledge but is itself constitutive of thinking and selfhood. Education should aim not just at producing correct answers but at drawing learners into genuine dialogue—with texts, with teachers and peers, with ideas themselves. Can AI play a legitimate role in such dialogue? This remains a genuinely open question.

Multiplex Systemics Analysis

The preceding analysis applies the framework's stratified levels (Figure 1.3) to a single technology, reading AI writing assistants across the technical, pedagogical, psychological, social, political-economic, and ethical-spiritual levels. A multiplex systemics approach recognizes that all these levels matter and interact in complex ways. Purely technical solutions (better AI) will not resolve pedagogical questions (how students learn to write). Pedagogical innovations must consider psychological impacts (on identity and agency), social contexts (norms of academic integrity), political-economic realities (access and equity), and ultimate purposes (wholeness-making versus fragmentation). Most existing discourse on AI writing assistants operates at only one or two levels, typically technical and pedagogical. This produces impoverished analysis, generating either naive enthusiasm ("AI will revolutionize writing instruction!") or blanket rejection ("AI will destroy writing!").

The wholeness-making orientation asks: Can we design AI writing tools and pedagogical practices that simultaneously support rather than replace authentic thinking (psychological level), enhance rather than bypass developmental processes (pedagogical level), foster rather than undermine authorial voice and agency (ethical level), reduce rather than exacerbate inequities (political level), and serve meaningful learning rather than mere efficiency (spiritual level)? This is not impossible, but it requires intentional design guided by normative commitment to *kamāl* rather than mere technological solutionism.

1.4.2 Design Principles Emerging from Multiplex Analysis

Based on this extended case analysis, we can articulate specific design principles for AI in education grounded in multiplex systemics.

Augment Rather Than Automate.

Design AI to augment human capabilities rather than automate human functions. This means enhancing teacher judgment rather than replacing it, supporting student agency rather than directing behavior, enabling richer human interaction rather than substituting for it, and providing

tools for thinking rather than doing thinking for students. The "centaur model" from chess—where human-AI teams outperform either alone—offers inspiration, though in education the human must remain primary, with AI as an instrument serving human purposes.

Make Systems Legible and Contestable.

Avoid black-box AI. Make systems understandable to teachers and students, provide mechanisms for human override and contestation, communicate clearly about capabilities and limitations, and teach critical understanding of how systems work. Legibility supports agency—the capacity to understand and potentially resist technological shaping.

Embed Technology in Rich Pedagogical Ecology.

Resist technological substitution and determinism. View AI as one element within larger pedagogical systems including direct teacher-student relationships, peer collaboration and dialogue, hands-on experiential learning, contemplative and creative practices, and connection to community and nature. No single approach suffices; wholeness requires diversity of pedagogical modes.

Attend to Multiple Dimensions.

Design with awareness of impact across all levels: Does it work reliably? (technical) Does it enhance learning? (pedagogical) Does it support healthy development? (psychological) Does it strengthen positive relationships? (social) Does it promote justice and equity? (political) Does it honor human dignity and transcendent meaning? (ethical-spiritual). Success at one level cannot justify harm at others.

Design for Polarity Management.

Build in mechanisms for dynamic balancing: switches between modes (AI assistance on/off), indicators showing when rebalancing is needed, flexibility for teachers to adjust based on context, and both automated elements and spaces for human judgment. Avoid rigid either-or architectures.

Orient Toward Meaningful Learning.

Ensure technology serves substantive educational purposes: deep understanding over superficial performance, transfer and application over narrow skills, curiosity and intrinsic motivation over compliance, wisdom and judgment over information processing, and character formation alongside cognitive growth. Resist the tyranny of the measurable.

1.5 Implications and Future Directions

1.5.1 Institutional and Policy Implications

Beyond specific technologies, wholeness-making requires institutional and policy changes. *First*, we must shift evaluation frameworks beyond narrow metrics (test scores, completion rates, efficiency gains) to holistic assessment encompassing student well-being and flourishing, development of agency and self-direction, quality of relationships and community, depth of understanding and wisdom, and ethical character and social responsibility. What gets measured shapes what gets valued; we must expand measurement to capture wholeness.

Second, we must resist technological determinism by countering the narrative that AI adoption is inevitable and education must "keep up." We should assert human agency and collective choice, recognizing that we can choose which technologies to adopt and how, that some educational goods may be incompatible with certain AI applications, that slower, more deliberate technological change may serve learning better, and that preservation of low-tech and no-tech spaces has value. Technology is a mean, not an end.

Third, rather than viewing AI as a substitute for human expertise, we must invest in human development through teacher training in learning sciences, ethics, and technology critique, professional development in integrative pedagogies, mental health support and counseling resources, and smaller class sizes enabling stronger relationships. The most important "technology" in education remains the skilled, caring human educator.

Fourth, we must encourage public deliberation by moving AI ethics from the technical domain to democratic discourse, engaging educators, students, families, and communities in decision-making, creating forums for reflection on values and purposes, ensuring diverse voices (especially marginalized communities), and building a shared understanding of complex trade-offs. Educational technology decisions are fundamentally political and ethical, requiring democratic legitimation.

Fifth, we should support alternative models that challenge monopolistic Silicon Valley approaches by supporting open-source educational AI, public educational technology infrastructure, research on human-centered and ethics-based design, and the sharing of best practices and critical perspectives. The current trajectory is not inevitable.

1.5.2 Contributions to Holistic Education Scholarship

This work contributes to broader scholarship on holistic education, joining voices calling for more integrated, humanistic approaches. Goldberg et al. (2014) argue for "a whole new engineer" prepared not just technically but in terms of analytical, creative, and professional skills integrated with an understanding of social context and human values. Perkins (2010) articulates seven principles for "making learning whole," emphasizing that effective learning requires understanding purposes, playing the whole game (not just isolated skills), making learning worth doing, working on hard parts, learning from teams and others, learning from inside (intrinsic motivation), and learning

from without (external feedback). Al Zeera (2001) presents an Islamic perspective on holistic education, arguing that wholeness and holiness are inseparable in authentic education—intellectual development must integrate with moral and spiritual formation.

Our multiplex systemics framework extends this scholarship by providing philosophical foundations for holistic education rooted in both Western and Islamic intellectual traditions, practical methodologies (polarity mapping, multi-level analysis) for navigating complexity, explicit attention to technology's role in education (which holistic education literature often underemphasizes), and normative orientation toward *kamāl* that goes beyond calls for "integration" to articulate substantive vision of human flourishing. Where previous holistic education scholarship has been strong on critique of reductionism but sometimes vague on alternatives, multiplex systemics offers concrete conceptual tools and analytical frameworks.

1.5.3 Limitations and Future Research Directions

Conceptually, our framework draws on Islamic philosophical concepts; as we noted at the outset, while these articulate a pluralistic insight we believe has universal applicability, the particular vocabulary may not resonate equally across cultural contexts. Future work should explore how multiplex systemics can be more fully translated into other philosophical traditions while maintaining integrity. Since abstract principles like "wholeness-making" and "kamāl" are difficult to operationalize into specific metrics, we need continued work on assessment approaches that capture multi-dimensional flourishing without reductive quantification.

Theoretically, we need bridges between multiplex systemics and other frameworks (capabilities approach, virtue ethics, indigenous knowledge systems), a more precise articulation of wholeness across developmental stages and cultural contexts, integration with critical perspectives on technology and power, and dialogue with empirical learning sciences. Finally, policy research should analyze governance models enabling democratic control over educational AI, regulatory frameworks incentivizing wholeness-making designs, economic models making ethical AI sustainable, and strategies for building public consciousness and mobilizing support for alternatives.

Practically, current institutional and market incentives favor optimization, efficiency, and scalability, not wholeness-making. Shifting these incentive structures requires not just better ideas but organized collective action. Moreover, multiplex analysis requires expertise across multiple domains; building this capacity in educational institutions is a long-term project. Finally, educational technology evolves rapidly while cultural and institutional change is slow; we need adaptive capacity to keep ethical frameworks relevant amid accelerating technological development.

Future research directions include empirical studies examining long-term impacts of educational AI on student development across multiple dimensions, the effectiveness of both-and designs compared to either-or approaches, factors enabling or hindering wholeness-oriented technology adoption, and cross-cultural variations in how multiplex principles are understood and applied. We also need design science research producing prototypes of wholeness-making educational AI systems,

participatory design processes involving educators and students, assessment instruments capturing multi-dimensional flourishing, and case studies of successful polarity management.

1.5.4 Conclusion: The Stakes of Our Technological Choices

We stand at a decisive moment in educational history. The integration of increasingly powerful AI into education is not simply a technical upgrade but a civilizational choice about what it means to be human and how we initiate the next generation into that humanity. The dominant paradigm of technological solutionism, guided by efficiency and optimization, promises quantifiable gains but threatens qualitative losses harder to see: atrophy of human agency, fragmentation of integrated selfhood, erosion of meaningful relationships, reduction of education to performance metrics, loss of connection to transcendent purpose. These losses represent movement away from *kamāl* toward *naqs*, not merely as individual deficiency but as collective spiritual impoverishment.

But this trajectory is not inevitable. As Holmes et al. (2025) argue, we need not be passive recipients of technological change. We can assert human agency and collective wisdom to shape technology according to our values rather than allowing technology to reshape us according to its logic. The multiplex systemics framework offers philosophical resources for this essential task. By recognizing the multi-layered nature of educational reality, attending to systemic dynamics and polarities, and orienting toward wholeness-making, we can design and deploy educational AI in ways that genuinely serve human flourishing.

This work extends our ongoing program of applied multiplexity (Qadir, 2022; Qadir and Senturk, 2026a; Qadir et al., 2025), demonstrating once more that multiplex thinking, which holds complexity while maintaining normative clarity, offers a path through contemporary challenges that neither naive optimism nor blanket rejection can provide. As we face an AI-mediated educational future, the question is not whether to use these powerful tools but how to use them wisely, in service of what ends, guided by which values. Multiplex systemics provides a conceptual foundation for engaging these questions with the seriousness they deserve.

The formation of persons, and the cultivation of wisdom, are too important to be left to algorithms optimized for engagement and efficiency. As educators, technologists, policymakers, and citizens, we face a choice: Will we allow educational technology to fragment and instrumentalize, or will we insist that it serve wholeness-making? Will we design AI systems that diminish human agency, or ones that augment human capacity for wisdom and virtue? Will we optimize for narrow metrics, or cultivate the fullness of human being?

These are not technical questions but profoundly moral and spiritual ones, demanding not merely clever solutions but ethical clarity, systems wisdom, and commitment to transcendent purposes. The transformation of educational technology must begin with the transformation of ourselves: our assumptions, our values, our vision of human possibility. May this work contribute to that necessary transformation, and to the emergence of educational futures serving not efficiency alone, but wisdom, justice, and human wholeness.

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